



**An Industry Oriented Mini Project (CS705PC)**

**on**

**Music Genre Classification**

**by**

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# ABSTRACT

Music genre classification addresses the challenge of categorizing songs into genres by analyzing their audio signals. While music perception is subjective and genre boundaries are often unclear, the development of automated systems helps overcome these complexities, making tasks like personalized playlists and music recommendations more efficient and user-friendly.

This project focuses on harnessing the power of Convolutional Neural Networks (CNNs) to extract meaningful features from spectrograms for accurate genre prediction. Beyond simplifying content organization, this approach contributes to improving recommendation engine functionality and ensures effective management of the ever-growing digital music landscape.

# INTRODUCTION

Music Genre Classification is a fundamental task in the field of music information retrieval. This project aims to classify audio tracks into one of ten genres: Blues, Classical, Country, Disco, Hip-Hop, Jazz, Metal, Pop, Reggae, and Rock. The GTZAN dataset, with 1000 audio tracks each 30 seconds long, sampled at 22050Hz, serves as the foundation for this classifier.

Our aim is to develop an effective music genre classification system using CNNs. With the proliferation of digital music, automated genre classification is essential for various applications. We'll detail our approach, including data preprocessing, model design, training, and evaluation, to create a robust system capable of accurately predicting music genres. Let's dive into our findings and insights.

# PROBLEM STATEMENT

The task of music genre classification involves analyzing audio signals and categorizing them into predefined genres such as rock, jazz, hip-hop, classical, etc. This task is challenging due to the complex and subjective nature of music perception, where genre boundaries can be ambiguous and subjective. However, automated genre classification is crucial for various applications such as music recommendation systems, content organization, and personalized playlists.

The problem to be addressed is to develop an accurate and efficient music genre classification system using machine learning techniques, particularly Convolutional Neural Networks (CNNs). Given a dataset of audio samples, the goal is to train a CNN model capable of accurately predicting the genre label of each audio sample. This involves preprocessing the audio data, extracting relevant features, designing and training a CNN architecture, and evaluating the model's performance in terms of classification accuracy, precision, recall, and F1-score across different music genres. The ultimate aim is to create a robust and scalable system that can classify music genres reliably across diverse datasets and real-world scenarios.

# MOTIVATION

The motivation for selecting "Music Genre Classification" as a mini-project lies in the increasing need for efficient organization and personalization in the rapidly expanding digital music industry. With vast music libraries and diverse listener preferences, an automated genre classification system simplifies navigation, enhances user experience in recommendation engines, and aids in managing large-scale music databases. By leveraging Convolutional Neural Networks (CNNs) to analyze audio features, this project combines cutting-edge technology with a practical application, offering valuable insights into music data while contributing to advancements in Music Information Retrieval (MIR).



# TECHNICAL SPECIFICATION

## Software Requirements

- *Operating System: Windows 10 (or later versions), macOS, or Linux*
- *Technology: Python 3.x*
- *Libraries and Frameworks:*
  - *TensorFlow and Keras: For building and training CNN models*
  - *Librosa: For audio processing and feature extraction*
  - *Pandas and NumPy: For data manipulation and analysis*
  - *Matplotlib: For visualizing results*
  - *Gradio: For creating an interactive user interface*
  - *Scikit-learn: For performance evaluation and data splitting*
- *IDE: Jupyter Notebook or Google Collab*

## Hardware Requirements

- *Processor: Intel Core i5 or equivalent*
- *RAM: Minimum 8 GB*
- *Storage: Minimum 250 GB (SSD recommended)*
- *Graphics Processing Unit (GPU): NVIDIA GPU with CUDA support (optional but recommended for faster training)*

# LITERATURE SURVEY

| S. No | Author                                                                                       | Title                                                     | Year | Journal                                                                     | Methodology                                                                                                                         | Merits                                                                                                                                                                    | Demerits                                                                                                                                                         |
|-------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------|------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Shajin Prince, Justin Jojoy Thomas, Sharon Jostana J, Kakarla Preethi Priya, J Joshua Daniel | Music Genre Classification using Deep Learning - A Review | 2023 | International Conference on Signal Processing, Communication and Networking | The paper reviews deep learning techniques for music genre classification, focusing on CNNs, RNNs, and hybrid models like CNN-RNNs. | <ul style="list-style-type: none"><li>- Comprehensive review of modern deep learning architectures.</li><li>- Discusses performance across different datasets..</li></ul> | <ul style="list-style-type: none"><li>- Limited experimental analysis.</li><li>- Primarily theoretical with limited real-world implementation results.</li></ul> |

# LITERATURE SURVEY

| S.No | Author                                                                                 | Title                                                                                                        | Year | Journal | Methodology                                                                                                                                                                                                                                                             | Merits                                                                                                                                                                               | Demerits                                                                                                                                                                |
|------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | Bishwomkar Pani<br>grahi, Varun<br>Gandhi, Rahul<br>Bhandari, Shivraj,<br>Kumari Priya | Harmony<br>in Algorithms:<br>Exploring Musi<br>c<br>Genre Classific<br>ation through<br>Machine Learni<br>ng | 2023 | IEEE    | The<br>methodology involves<br>using the GTZAN<br>dataset, extracting<br>spectral features,<br>applying a CNN<br>for classification,<br>and comparing it<br>with SVM, KNN,<br>Naive Bayes, and<br>Random Forest for<br>accuracy and<br>validation<br>loss minimization. | CNN<br>achieved high<br>accuracy of<br>79%, outperfor<br>med traditional<br>methods (SVM,<br>KNN, Naive<br>Bayes, Random<br>Forest). Reduce<br>d validation loss<br>es significantly | Limited to a single<br>dataset (GTZAN),<br>may not<br>generalize<br>to other<br>music genres<br>or datasets.<br>Other models<br>achieved lower<br>accuracy than<br>CNN. |



# LITERATURE SURVEY

| S.No | Author                           | Title                                     | Year | Journal                                                          | Methodology                                                                                             | Merits                                                                                                                                     | Demerits                                                                                                     |
|------|----------------------------------|-------------------------------------------|------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 3    | Nikki Pelchat, Craig M. Gelowitz | Neural Network Music Genre Classification | 2020 | CANADIAN JOURNAL OF ELECTRICAL AND COMPUTER ENGINEERING, VOL. 43 | Utilized neural networks (NNs) with spectrogram images as inputs to classify songs into musical genres. | Effective use of spectrograms to capture time-frequency representations of audio. Explores various NN techniques for music classification. | Limited success across different libraries. Highly dependent on the quality and preprocessing of input data. |

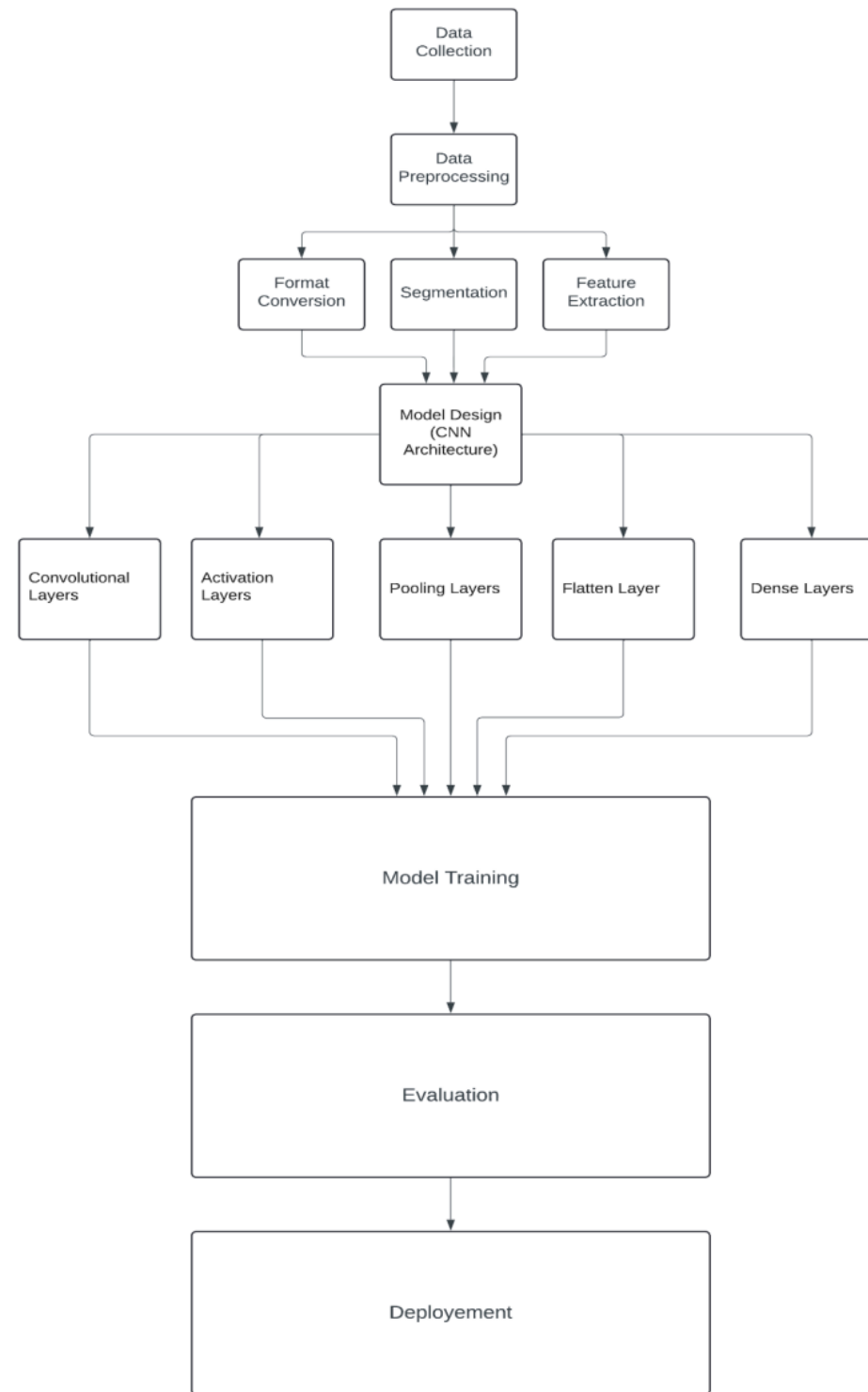
# LITERATURE SURVEY

| S.No | Author                                                       | Title                                                       | Year | Journal          | Methodology                                                                                                                              | Merits                                                                                                                                                           | Demerits                                                                                                                                     |
|------|--------------------------------------------------------------|-------------------------------------------------------------|------|------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 4    | Mohsin Ashraf, Ikram Ud Din, Jawad Rasheed, Mirsat Yesiltepe | A Hybrid CNN and RNN Variant Model for Music Classification | 2023 | Applied Sciences | Combines Convolutional Neural Networks (CNN) with Recurrent Neural Networks (RNN) to handle spatial and temporal patterns in audio data. | <ul style="list-style-type: none"><li>- Effectively handles sequential and non-sequential audio features.</li><li>- High accuracy on complex datasets.</li></ul> | <ul style="list-style-type: none"><li>- Requires significant computational power.</li><li>- Prone to overfitting on small datasets</li></ul> |

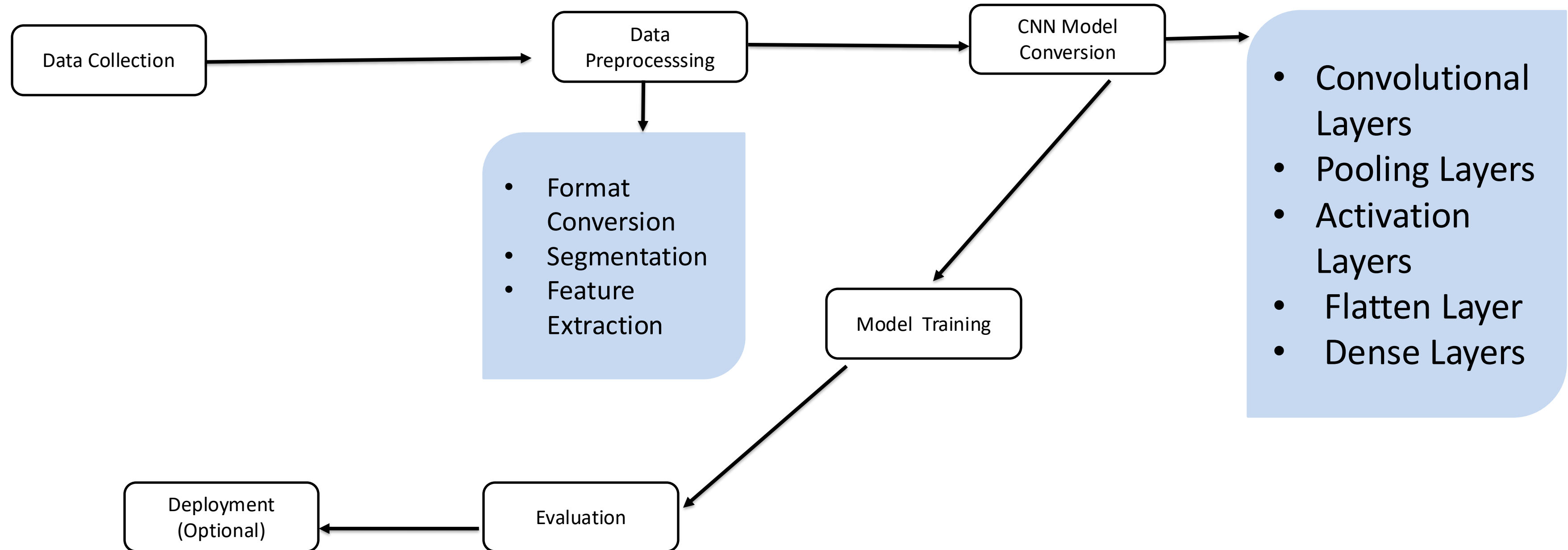
# LITERATURE SURVEY

| S.No | Author                                                                         | Title                                                                                     | Year | Journal      | Methodology                                                                                                          | Merits                                                                                                                                                         | Demerits                                                                                                                                              |
|------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------|--------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5    | <a href="#">Aru Upadhyay, Anusha Barman, Disha Gupta &amp; Dhirendra Kumar</a> | Exploring Empirical Mode Decomposition for Music Genre Classification Using Deep Learning | 2023 | SpringerLink | Empirical Mode Decomposition (EMD) with deep learning models to improve feature extraction for genre classification. | <ul style="list-style-type: none"><li>- Enhanced feature extraction using EMD.</li><li>- Improved classification accuracy for complex audio signals.</li></ul> | <ul style="list-style-type: none"><li>- Computationally intensive.</li><li>- Performance highly dependent on the quality of input features.</li></ul> |

# DESIGN METHODOLOGY



# Block Diagram





# IMPLEMENTATION

## **Data Collection:**

- Gathering a dataset of audio samples labeled with their corresponding genres. The dataset should be diverse and encompass various music styles.

## **Data Processing:** Focuses on preparing the audio data for the CNN model.

- **Format Conversion:** Ensure all audio files have a consistent format (e.g., sampling rate, bit depth). Based on this i want architecture diagram in 2d for music genre classifier
- **Segmentation:** Divide audio files into shorter segments (frames) for analysis.
- **Feature Extraction:** Extract relevant features from the audio segments. These features could include Mel-Frequency Cepstral Coefficients (MFCCs), Spectral Flux, and Chroma features, which capture aspects of rhythm, timbre, and harmony.

# IMPLEMENTATION

**Model Design (CNN Architecture):** Focuses on designing the CNN architecture for genre classification. The architecture typically includes:

- **Convolutional Layers:** These layers extract features from the audio data using learnable filters.
- **Pooling Layers:** These layers down sample the data, reducing complexity and potentially improving generalization.
- **Activation Layers:** These layers introduce non-linearity into the model, allowing it to learn complex relationships.
- **Flatten Layer:** This layer transforms the multi-dimensional output from convolutional layers into a one-dimensional vector suitable for classification.
- **Dense Layers:** These fully-connected layers perform classification based on the extracted features. The final layer has an output size equal to the number of genres to predict.

# IMPLEMENTATION

- **Model Training:**

Training the CNN model using the prepared dataset. The training process optimizes the model's weights and biases to minimize the classification error for the given genres.

- **Evaluation:**

Focuses on assessing the model's performance on unseen data.

- **Deployment (Optional)**

Integrating the trained model into a real-world application, such as a music streaming service, for automatic genre classification of new audio data.

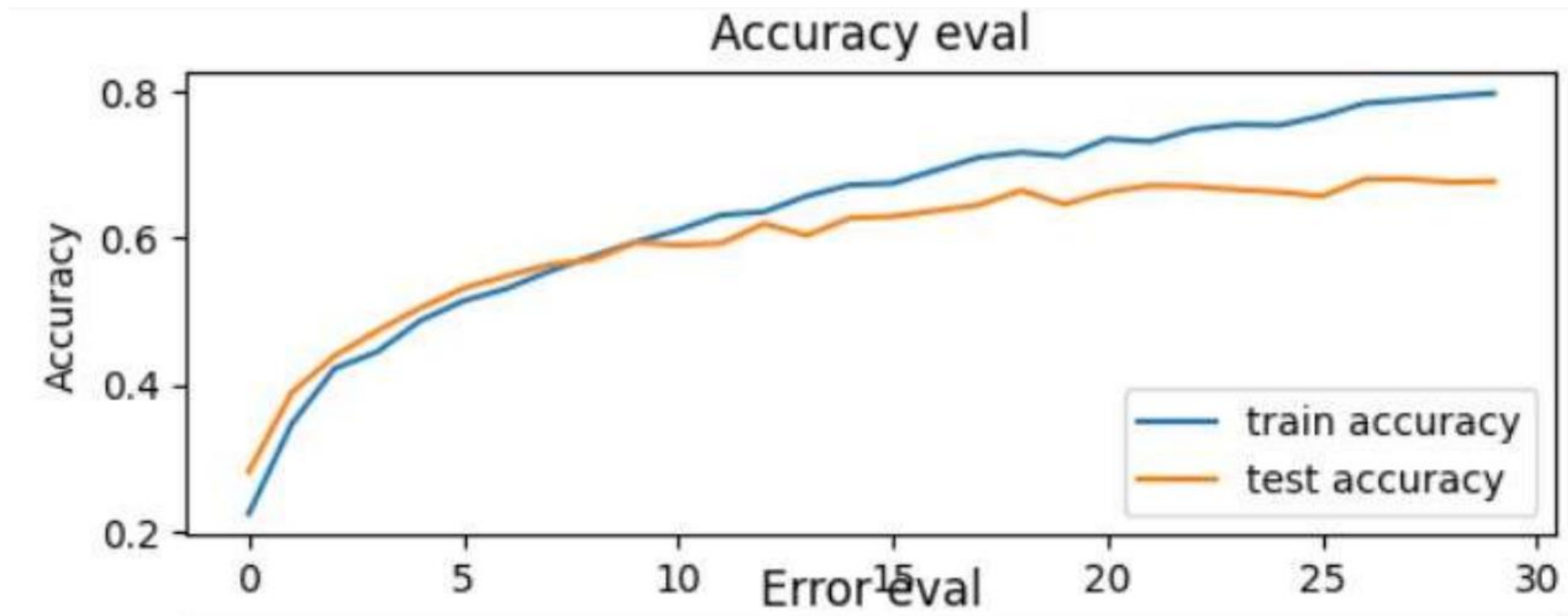


# RESULTS

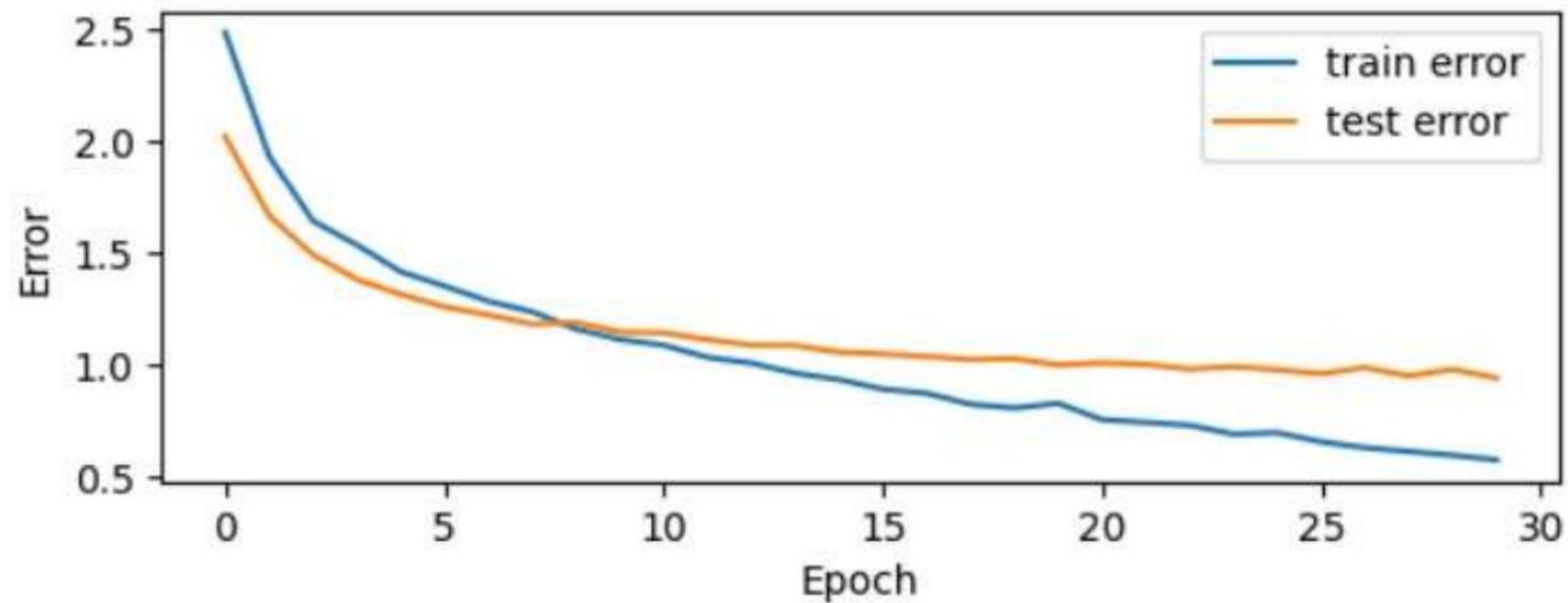
Model: "sequential"

| Layer (type)                                | Output Shape        | Param # |
|---------------------------------------------|---------------------|---------|
| conv2d (Conv2D)                             | (None, 214, 11, 32) | 320     |
| max_pooling2d (MaxPooling2D)                | (None, 107, 6, 32)  | 0       |
| batch_normalization (Batch Normalization)   | (None, 107, 6, 32)  | 128     |
| conv2d_1 (Conv2D)                           | (None, 105, 4, 32)  | 9248    |
| max_pooling2d_1 (MaxPooling2D)              | (None, 53, 2, 32)   | 0       |
| batch_normalization_1 (Batch Normalization) | (None, 53, 2, 32)   | 128     |
| conv2d_2 (Conv2D)                           | (None, 52, 1, 32)   | 4128    |
| max_pooling2d_2 (MaxPooling2D)              | (None, 26, 1, 32)   | 0       |
| batch_normalization_2 (Batch Normalization) | (None, 26, 1, 32)   | 128     |
| flatten (Flatten)                           | (None, 832)         | 0       |
| dense (Dense)                               | (None, 64)          | 53312   |
| dropout (Dropout)                           | (None, 64)          | 0       |
| dense_1 (Dense)                             | (None, 10)          | 650     |

## Accuracy vs Epoch plot:



## Loss vs Epoch plot:



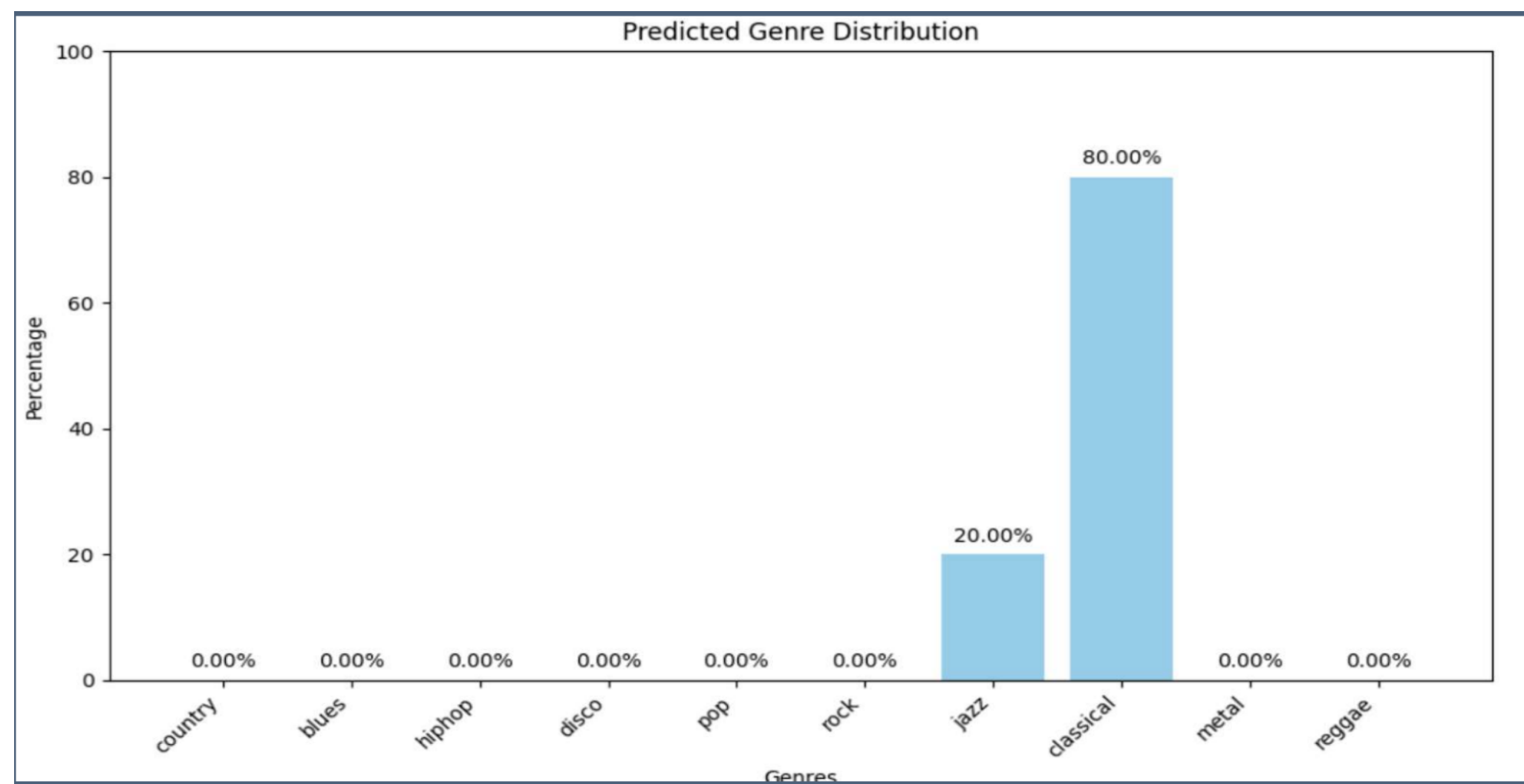
# Output:

Upload an audio file and get the predicted genre along with genre distribution percentages.

audio\_file

day dreaming.wav 5.0 MB ↓

Clear Submit



# FUTURE SCOPE

- Dataset Expansion and Diversity:** Include diverse genres and multilingual tracks for improved accuracy and reduced bias.
- Incorporating Additional Features:** Use tempo, rhythm, and lyrics to capture richer musical characteristics.
- Edge Computing for On-device Classification:** Enable real-time genre detection without internet dependency for seamless use.
- Real-Time Classification and Streaming Integration:** Integrate with apps for instant tagging and personalized playlist generation.

# CONCLUSION

In summary, the project successfully built a music genre classifier using machine learning. It utilized MFCC features and a CNN model to predict genres with reasonable accuracy. The integration of Gradio provided a user-friendly interface for easy interaction. Overall, the project showcased the potential of ML in music analysis, with room for future enhancements in model refinement and UI improvements.



# REFERENCES

- [1] G. Tzanetakis and P. Cook. Musical genre classification of audio signals. *Speech and Audio Processing, IEEE Transactions on*, 10(5):293–302, Jul 2002.
- [2] N. Scaringella, G. Zoia, and D. Mlynek, —Automatic genre classification of music content: a survey‖, *IEEE Signal Processing Magazine*, Vol. 23, Issue 2, pp. 133–141, 2006.
- [3] <https://www.kaggle.com/datasets/andradaolteanu/gtzan-dataset-music-genre-classification>

**THANK YOU**